NETWORKING HOME LAB PORTFOLIO

Hands-on configuration • Verification • Troubleshooting

Project 1	MikroTik Routing and VLAN Lab
Project 2	Ubuntu Server and WireGuard VPN
Project 3	UniFi Wireless Network Integration
Project 4	Cisco Packet Tracer Networking Labs

Prepared as a practical learning portfolio for entry-level Network Technician, IT Support, and Junior/Associate Network Engineer opportunities.

This document records practical lab work and the troubleshooting methods used. It distinguishes confirmed observations from incomplete work.

Project 1 — MikroTik Routing and VLAN Lab

Physical networking home lab | MikroTik CloudCoreRouter1036, CloudRouterSwitch326, hEX and hAP devices

1. Project Overview

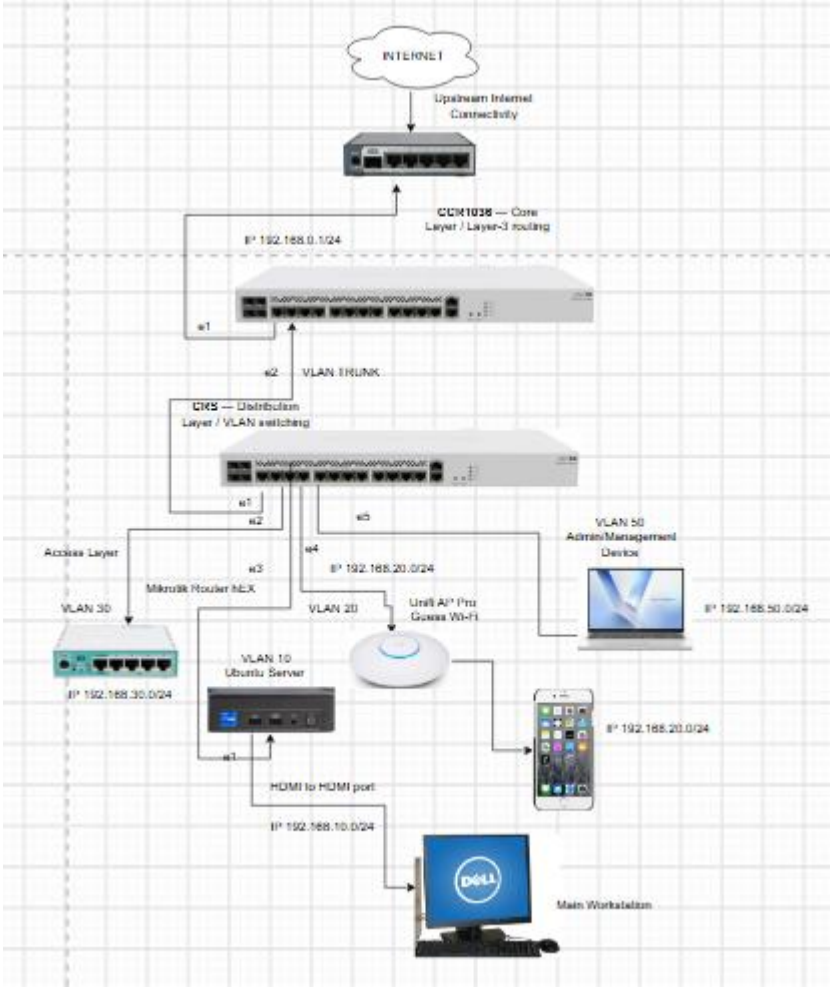
This project documents hands-on practice with MikroTik routers and switches to learn LAN addressing, gateway configuration, NAT, VLAN concepts, port membership, and connectivity verification. The lab has been developed in stages, and the full multi-VLAN topology should be treated as a work in progress until every VLAN has been tested end-to-end.

2. Objectives

- Review device roles: core routing, distribution switching, and access connectivity.
- Configure and inspect IPv4 addressing, gateway settings, DHCP-related settings, and NAT.
- Practice VLAN interfaces, bridge VLAN tables, tagged and untagged ports, and VLAN filtering.
- Verify connectivity methodically and document faults, changes, and test results.

3. Network Topology

Internet/WAN → hAP ac lite upstream→ MikroTik core router (CCR1036, planned core role) → CRS326 distribution switch → Mikrotik Router hEX access devices → PCs, Intel NUC server, and UniFi APs.



4. Configuration and Commands

The following are examples/reference commands. Adapt interface names, IPs, VLAN IDs, and device versions to the configuration you actually tested.

Useful inspection commands

```
/interface print detail
/ip address print detail
/ip route print detail
/ip dhcp-client print detail
/ip firewall nat print detail
/interface bridge print detail
/interface bridge vlan print detail
```

Example source NAT rule (adapt interface name)

```
/ip firewall nat add chain=srcnat out-interface=<WAN_INTERFACE> action=masquerade
```

Example VLAN interface (adapt parent interface and VLAN ID)

```
/interface vlan add name=vlan10 vlan-id=10 interface=<PARENT_INTERFACE>
```

5. Verification and Testing

- Confirm physical link state and interface names before changing configuration.

```
[admin@MikroTik] > /interface print
Flags: R - RUNNING; S - SLAVE
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAX-L2MTU, MAC-ADDRESS
 #    NAME         TYPE    ACTUAL-MTU  L2MTU  MAX-L2MTU  MAC-ADDRESS
 0 R  ether1       ether         1500   1580      10222  CC:2D:E0:BB:AB:79
 1    ether2       ether         1500   1580      10222  CC:2D:E0:BB:AB:7A
 2 RS ether3       ether         1500   1580      10222  CC:2D:E0:BB:AB:7B
 3 RS ether4       ether         1500   1580      10222  CC:2D:E0:BB:AB:7C
 4 RS ether5       ether         1500   1580      10222  CC:2D:E0:BB:AB:7D
 5  S ether6       ether         1500   1580      10222  CC:2D:E0:BB:AB:7E
 6  S ether7       ether         1500   1580      10222  CC:2D:E0:BB:AB:7F
 7  S ether8       ether         1500   1580      10222  CC:2D:E0:BB:AB:80
 8  S ether9       ether         1500   1580      10222  CC:2D:E0:BB:AB:81
 9  S ether10      ether         1500   1580      10222  CC:2D:E0:BB:AB:82
10  S ether11      ether         1500   1580      10222  CC:2D:E0:BB:AB:83
11  S ether12      ether         1500   1580      10222  CC:2D:E0:BB:AB:84
12  S sfp1         ether         1500   1580      10222  CC:2D:E0:BB:AB:75
13  S sfp2         ether         1500   1580      10222  CC:2D:E0:BB:AB:76
14  S sfp3         ether         1500   1580      10222  CC:2D:E0:BB:AB:77
15  S sfp4         ether         1500   1580      10222  CC:2D:E0:BB:AB:78
;;; defconf
16 R  CloudRouter  bridge        1500   1580             CC:2D:E0:BB:AB:79
17 R  VLAN10       vlan          1500   1576             CC:2D:E0:BB:AB:79
18 R  VLAN20       vlan          1500   1576             CC:2D:E0:BB:AB:79
19 R  VLAN30       vlan          1500   1576             CC:2D:E0:BB:AB:79
20    VLAN40       vlan          1500   1576             CC:2D:E0:BB:AB:7A
21 R  VLAN50       vlan          1500   1576             CC:2D:E0:BB:AB:79
```

```
[admin@MikroTik] > /interface ethernet print
Flags: R - RUNNING; S - SLAVE
Columns: NAME, MTU, MAC-ADDRESS, ARP
 #    NAME      MTU  MAC-ADDRESS        ARP
 0 R  ether1   1500  CC:2D:E0:BB:AB:79  enabled
 1    ether2   1500  CC:2D:E0:BB:AB:7A  enabled
 2 RS ether3   1500  CC:2D:E0:BB:AB:7B  enabled
 3 RS ether4   1500  CC:2D:E0:BB:AB:7C  enabled
 4 RS ether5   1500  CC:2D:E0:BB:AB:7D  enabled
 5  S ether6   1500  CC:2D:E0:BB:AB:7E  enabled
 6  S ether7   1500  CC:2D:E0:BB:AB:7F  enabled
 7  S ether8   1500  CC:2D:E0:BB:AB:80  enabled
 8  S ether9   1500  CC:2D:E0:BB:AB:81  enabled
 9  S ether10  1500  CC:2D:E0:BB:AB:82  enabled
10  S ether11  1500  CC:2D:E0:BB:AB:83  enabled
11  S ether12  1500  CC:2D:E0:BB:AB:84  enabled
12  S sfp1     1500  CC:2D:E0:BB:AB:75  enabled
13  S sfp2     1500  CC:2D:E0:BB:AB:76  enabled
14  S sfp3     1500  CC:2D:E0:BB:AB:77  enabled
15  S sfp4     1500  CC:2D:E0:BB:AB:78  enabled
```

```
[admin@MikroTik] > /interface ethernet monitor ether1 once
                     name: ether1
                   status: link-ok
          auto-negotiation: done
                     rate: 100Mbps
              full-duplex: yes
           tx-flow-control: no
           rx-flow-control: no
              advertising: 10M-full,100M-full,1000M-full
  link-partner-advertising: 10M-half,10M-full,100M-half,100M-full
[admin@MikroTik] > /interface ethernet monitor ether2 once
                     name: ether2
                   status: no-link
          auto-negotiation: done
              advertising: 10M-full,100M-full,1000M-full
  link-partner-advertising:
[admin@MikroTik] > /interface ethernet monitor ether3 once
                     name: ether3
                   status: link-ok
          auto-negotiation: done
                     rate: 1Gbps
              full-duplex: yes
           tx-flow-control: no
           rx-flow-control: no
              advertising: 10M-full,100M-full,1000M-full
  link-partner-advertising: 10M-half,10M-full,100M-half,100M-full,1000M-full
[admin@MikroTik] > /interface ethernet monitor ether4 once
                     name: ether4
                   status: link-ok
          auto-negotiation: done
                     rate: 1Gbps
              full-duplex: yes
           tx-flow-control: no
           rx-flow-control: no
              advertising: 10M-full,100M-full,1000M-full
  link-partner-advertising: 10M-half,10M-full,100M-half,100M-full,1000M-full
[admin@MikroTik] > /interface ethernet monitor ether5 once
                     name: ether5
                   status: link-ok
          auto-negotiation: done
```

- Check IP addresses, default route, DHCP client/server state, and NAT rules.

```
[admin@MikroTik] > /ip address print detail
Flags: X - disabled, I - invalid, D - dynamic
 0   address=192.168.10.1/24 network=192.168.10.0 interface=VLAN10 actual-interface=VLAN10

 1   address=192.168.20.1/24 network=192.168.20.0 interface=VLAN20 actual-interface=VLAN20

 2   address=192.168.30.1/24 network=192.168.30.0 interface=VLAN30 actual-interface=VLAN30

 3   address=192.168.40.1/24 network=192.168.40.0 interface=VLAN40 actual-interface=VLAN40

 4   address=192.168.50.1/24 network=192.168.50.0 interface=VLAN50 actual-interface=VLAN50

 5 D address=192.168.110.254/24 network=192.168.110.0 interface=ether1 actual-interface=ether1
```

```
[admin@MikroTik] > /ip route print detail
Flags: D - dynamic; X - disabled, I - inactive, A - active; c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - vpn, m - modem, y - bgp-mpls-vpn;
H - hw-offloaded; + - ecmp
 DAd  dst-address=0.0.0.0/0 routing-table=main pref-src="" gateway=192.168.110.1 immediate-gw=192.168.110.1%ether1 distance=1 scope=30 target-scope=10 vrf-interface=ether1
      suppress-hw-offload=no

 DAc  dst-address=192.168.10.0/24 routing-table=main gateway=VLAN10 immediate-gw=VLAN10 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.10.1%VLAN10

 DAc  dst-address=192.168.20.0/24 routing-table=main gateway=VLAN20 immediate-gw=VLAN20 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.20.1%VLAN20

 DAc  dst-address=192.168.30.0/24 routing-table=main gateway=VLAN30 immediate-gw=VLAN30 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.30.1%VLAN30

 DIcH dst-address=192.168.40.0/24 routing-table=main gateway=VLAN40 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.40.1%VLAN40

 DAc  dst-address=192.168.50.0/24 routing-table=main gateway=VLAN50 immediate-gw=VLAN50 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.50.1%VLAN50
```

Activate Windows

```
[admin@MikroTik] > /ip dns print
                    servers:
              dynamic-servers: 192.168.110.1
               use-doh-server:
              verify-doh-cert: no
 doh-max-server-connections: 5
 doh-max-concurrent-queries: 50
                 doh-timeout: 5s
      allow-remote-requests: no
        max-udp-packet-size: 4096
       query-server-timeout: 2s
        query-total-timeout: 10s
     max-concurrent-queries: 100
max-concurrent-tcp-sessions: 20
                 cache-size: 2048KiB
              cache-max-ttl: 1w
    address-list-extra-time: 0s
                 cache-used: 28KiB
[admin@MikroTik] >
```

```
[admin@MikroTik] > /ping 8.8.8.8 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 112 19ms701us
    1 8.8.8.8                                    56 112 12ms791us
    2 8.8.8.8                                    56 112 14ms172us
    3 8.8.8.8                                    56 112 13ms606us
    4 8.8.8.8                                    56 112 9ms961us
    sent=5 received=5 packet-loss=0% min-rtt=9ms961us avg-rtt=14ms46us max-rtt=19ms701us

[admin@MikroTik] > /ping google.com count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 172.217.27.78                              56 112 16ms261us
    1 172.217.27.78                              56 112 13ms834us
    2 172.217.27.78                              56 112 14ms738us
    3 172.217.27.78                              56 112 11ms148us
    4 172.217.27.78                              56 112 15ms816us
    sent=5 received=5 packet-loss=0% min-rtt=11ms148us avg-rtt=14ms359us max-rtt=16ms261us
```

```
[admin@MikroTik] > /ip dhcp-client print detail
Flags: X - disabled, I - invalid, D - dynamic
 0   interface=ether1 add-default-route=yes default-route-distance=1 use-peer-dns=yes use-peer-ntp=yes dhcp-options=hostname,clientid status=bound address=192.168.110.254/24
     gateway=192.168.110.1 dhcp-server=192.168.88.251 primary-dns=192.168.110.1 expires-after=6m32s
[admin@MikroTik] > /ip dhcp-server print detail
Flags: D - dynamic; X - disabled, I - invalid
 0    name="dhcp10" interface=VLAN10 lease-time=30m address-pool=pool10 authoritative=yes use-radius=no lease-script=""

 1    name="dhcp20" interface=VLAN20 lease-time=30m address-pool=pool20 authoritative=yes use-radius=no lease-script=""

 2    name="dhcp30" interface=VLAN30 lease-time=30m address-pool=pool30 authoritative=yes use-radius=no lease-script=""

 3  I name="dhcp40" interface=VLAN40 lease-time=30m address-pool=pool40 authoritative=yes use-radius=no lease-script=""

 4    name="dhcp50" interface=VLAN50 lease-time=30m address-pool=pool50 authoritative=yes use-radius=no lease-script=""
[admin@MikroTik] > /ip dhcp-server lease print
Flags: D - DYNAMIC
Columns: ADDRESS, MAC-ADDRESS, HOST-NAME, SERVER, STATUS, LAST-SEEN
#   ADDRESS         MAC-ADDRESS        HOST-NAME         SERVER  STATUS  LAST-SEEN
0 D 192.168.10.200  94:C6:91:11:DB:7A  ubuntuserver      dhcp10  bound   13m29s
1 D 192.168.50.200  14:DA:E9:66:CC:A2  DESKTOP-C7HCUCL   dhcp50  bound   13m29s
2 D 192.168.20.198  EE:B6:DE:54:C6:DC  realme-C12        dhcp20  bound   1h41m10s
3 D 192.168.20.199  80:2A:A8:C6:AD:EF                    dhcp20  bound   11m
[admin@MikroTik] >
```

Activate Windows
Go to Settings to activate Windo

```
[admin@MikroTik] > /ip firewall nat print detail
Flags: X - disabled, I - invalid; D - dynamic
 0    ;;; Internet
      chain=srcnat action=masquerade out-interface=ether1
```

- Inspect Bridge → VLANs and confirm intended tagged/untagged membership.

```
[admin@MikroTik] > /interface bridge print detail
Flags: X - disabled, R - running
 0 R ;;; defconf
     name="CloudRouter" mtu=auto actual-mtu=1500 l2mtu=1580 arp=enabled arp-timeout=auto mac-address=CC:2D:E0:BB:AB:79 protocol-mode=rstp fast-forward=yes igmp-snooping=no
     auto-mac=no admin-mac=CC:2D:E0:BB:AB:79 ageing-time=5m priority=0x8000 max-message-age=20s forward-delay=15s transmit-hold-count=6 vlan-filtering=yes ether-type=0x8100 pvid=1
     frame-types=admit-all ingress-filtering=yes dhcp-snooping=no
[admin@MikroTik] > /interface bridge port print detail
Flags: X - disabled, I - inactive; D - dynamic; H - hw-offload
 0     ;;; defconf
       interface=ether3 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=10 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 1     ;;; defconf
       interface=ether4 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=20 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 2     ;;; defconf
       interface=ether5 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=50 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 3 I   ;;; defconf
       interface=ether6 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=30 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 4 I   ;;; defconf
       interface=ether7 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=1 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 5 I   ;;; defconf
       interface=ether8 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=1 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no

 6 I   ;;; defconf
       interface=ether9 bridge=CloudRouter priority=0x80 path-cost=10 internal-path-cost=10 edge=auto point-to-point=auto learn=auto horizon=none hw=yes auto-isolate=no
       restricted-role=no restricted-tcn=no pvid=1 frame-types=admit-all ingress-filtering=yes unknown-unicast-flood=yes unknown-multicast-flood=yes broadcast-flood=yes
       tag-stacking=no bpdu-guard=no trusted=no multicast-router=temporary-query fast-leave=no
```

```
[admin@MikroTik] > /interface bridge vlan print detail
Flags: X - disabled, D - dynamic
 0   bridge=CloudRouter vlan-ids=10 tagged=CloudRouter untagged

 1   bridge=CloudRouter vlan-ids=20 tagged=CloudRouter untagged

 2   bridge=CloudRouter vlan-ids=50 tagged=CloudRouter untagged

 3   bridge=CloudRouter vlan-ids=30 tagged=CloudRouter untagged

 4 D bridge=CloudRouter vlan-ids=1 tagged="" untagged="" curren
[admin@MikroTik] > /interface vlan print detail
Flags: X - disabled, R - running
 0 R name="VLAN10" mtu=1500 l2mtu=1576 mac-address=CC:2D:E0:BB:
     loop-protect-disable-time=5m vlan-id=10 interface=CloudRou

 1 R name="VLAN20" mtu=1500 l2mtu=1576 mac-address=CC:2D:E0:BB:
     loop-protect-disable-time=5m vlan-id=20 interface=CloudRou
```

- Test same-VLAN connectivity first, then gateway reachability, inter-VLAN routing if configured, and Internet access.

```
[admin@MikroTik] > /ping 192.168.20.199 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.20.199                             56  64 375us
    1 192.168.20.199                             56  64 540us
    2 192.168.20.199                             56  64 329us
    sent=3 received=3 packet-loss=0% min-rtt=329us avg-rtt=414us max-rtt=540us

[admin@MikroTik] > /ping 192.168.10.200 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.10.200                             56  64 813us
    1 192.168.10.200                             56  64 696us
    2 192.168.10.200                             56  64 925us
    3 192.168.10.200                             56  64 697us
    4 192.168.10.200                             56  64 722us
    sent=5 received=5 packet-loss=0% min-rtt=696us avg-rtt=770us max-rtt=925us

[admin@MikroTik] > /ping 192.168.20.199 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.20.199                             56  64 553us
    1 192.168.20.199                             56  64 535us
    2 192.168.20.199                             56  64 544us
    3 192.168.20.199                             56  64 300us
    4 192.168.20.199                             56  64 321us
    sent=5 received=5 packet-loss=0% min-rtt=300us avg-rtt=450us max-rtt=553us

[admin@MikroTik] > /ping 192.168.50.200 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.50.200                             56 128 4ms341us
    1 192.168.50.200                             56 128 587us
    2 192.168.50.200                             56 128 623us
    3 192.168.50.200                             56 128 9ms674us
    4 192.168.50.200                             56 128 1ms29us
    sent=5 received=5 packet-loss=0% min-rtt=587us avg-rtt=3ms250us max-rtt=9ms674us
```

```
[admin@MikroTik] > /ping 192.168.50.200 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.50.200                             56 128 4ms341us
    1 192.168.50.200                             56 128 587us
    2 192.168.50.200                             56 128 623us
    3 192.168.50.200                             56 128 9ms674us
    4 192.168.50.200                             56 128 1ms29us
    sent=5 received=5 packet-loss=0% min-rtt=587us avg-rtt=3ms250us max-rtt=9ms674us

[admin@MikroTik] > /ping 192.168.10.1 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.10.1                               56  64 208us
    1 192.168.10.1                               56  64 167us
    2 192.168.10.1                               56  64 186us
    3 192.168.10.1                               56  64 166us
    4 192.168.10.1                               56  64 173us
    sent=5 received=5 packet-loss=0% min-rtt=166us avg-rtt=180us max-rtt=208us

[admin@MikroTik] > /ping 192.168.20.1 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.20.1                               56  64 204us
    1 192.168.20.1                               56  64 166us
    2 192.168.20.1                               56  64 183us
    3 192.168.20.1                               56  64 177us
    4 192.168.20.1                               56  64 184us
    sent=5 received=5 packet-loss=0% min-rtt=166us avg-rtt=182us max-rtt=204us

[admin@MikroTik] > /ping 192.168.50.1 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.50.1                               56  64 204us
    1 192.168.50.1                               56  64 186us
    2 192.168.50.1                               56  64 188us
    3 192.168.50.1                               56  64 196us
    4 192.168.50.1                               56  64 169us
    sent=5 received=5 packet-loss=0% min-rtt=169us avg-rtt=188us max-rtt=204us

[admin@MikroTik] > /ping 192.168.40.1 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.40.1                               56  64 215us
    1 192.168.40.1                               56  64 192us

[admin@MikroTik] > /ping 8.8.8.8 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 8.8.8.8                                    56 112 24ms761us
    1 8.8.8.8                                    56 112 13ms12us
    2 8.8.8.8                                    56 112 20ms864us
    3 8.8.8.8                                    56 112 16ms450us
    4 8.8.8.8                                    56 112 10ms312us
    sent=5 received=5 packet-loss=0% min-rtt=10ms312us avg-rtt=17ms79us max-rtt=24ms761us

[admin@MikroTik] > /ping google.com count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 142.251.170.113                            56 103 32ms875us
    1 142.251.170.113                            56 103 33ms522us
    2 142.251.170.113                            56 103 35ms916us
    3 142.251.170.113                            56 103 29ms214us
    4 142.251.170.113                            56 103 31ms934us
    sent=5 received=5 packet-loss=0% min-rtt=29ms214us avg-rtt=32ms692us max-rtt=35ms916us
```

```
[admin@MikroTik] > /ping 192.168.20.198 count=5
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.20.198                             56  64 1ms493us
    1 192.168.20.198                             56  64 116ms690us
    2 192.168.20.198                             56  64 11ms19us
    3 192.168.20.198                             56  64 41ms949us
    4 192.168.20.198                             56  64 61ms274us
    sent=5 received=5 packet-loss=0% min-rtt=1ms493us avg-rtt=46ms485us max-rtt=116ms690us
```

6. Troubleshooting Experience

- Internet access failed because of an incorrectly configured NAT rule; after correcting the rule, external IP and domain connectivity tests succeeded in that troubleshooting session.

```
[admin@MikroTik] > /ip arp print detail
Flags: X - disabled, I - invalid, H - DHCP, D - dynamic, P - published; C - complete
 0 DC address=169.254.56.60 mac-address=14:DA:E9:66:CC:A2 interface=VLAN50 published=no

 1 DC address=192.168.20.199 mac-address=80:2A:A8:C6:AD:EF interface=VLAN20 published=no

 2 DC address=192.168.110.1 mac-address=C4:AD:34:05:B5:20 interface=ether1 published=no

 3 DC address=192.168.20.198 mac-address=EE:B6:DE:54:C6:DC interface=VLAN20 published=no

 4 DC address=192.168.50.200 mac-address=14:DA:E9:66:CC:A2 interface=VLAN50 published=no

 5 DC address=192.168.10.200 mac-address=94:C6:91:11:DB:7A interface=VLAN10 published=no
```

```
[admin@MikroTik] > /ip neighbor print detail
 0 interface=ether1 address=192.168.88.251 address4=192.168.88.251 mac-address=C4:AD:34:05:B5:21 identity="MikroTik" platform="MikroTik" version="7.24.4 (stable) 2026-09-16 11:32:21"
   unpack=none age=1s uptime=9h37m54s software-id="YI9B-WJX7" board="RB952Ui-5ac2nD" ipv6=no interface-name="bridge-LAN/ether4" system-caps="" system-caps-enabled=""
   discovered-by=cdp,mndp

 1 interface=ether4,CloudRouter address=192.168.20.199 address4=192.168.20.199 address6=fe80::822a:a8ff:fec6:adef mac-address=80:2A:A8:C6:AD:EF identity="UAP-AC-Pro-Gen2" platform=""
   version="" unpack=none age=13s ipv6=yes interface-name="eth0" system-description="UAP-AC-Pro-Gen2, 6.8.2.15592" system-caps=bridge,wlan-ap,router,station-only
   system-caps-enabled=bridge,wlan-ap,router discovered-by=lldp

 2 interface=ether5,CloudRouter mac-address=14:DA:E9:66:CC:A2 identity="" platform="" version="" unpack=none age=2m42s system-caps="" system-caps-enabled="" discovered-by=lldp
[admin@MikroTik] >
```

Activate Windows
Go to Settings to activate Windows.

- Inspected bridge VLAN tables, port assignments, VLAN interfaces, and VLAN-filtering status while diagnosing VLAN connectivity.

```
[admin@MikroTik] > /ip route print detail
Flags: D - dynamic; X - disabled, I - inactive, A - active; c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - vpn, m - modem, y - bgp-mpls-vpn;
H - hw-offloaded; + - ecmp
   DAd   dst-address=0.0.0.0/0 routing-table=main pref-src="" gateway=192.168.110.1 immediate-gw=192.168.110.1%ether1 distance=1 scope=30 target-scope=10 vrf-interface=ether1
         suppress-hw-offload=no

   DAc   dst-address=192.168.10.0/24 routing-table=main gateway=VLAN10 immediate-gw=VLAN10 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.10.1%VLAN10

   DAc   dst-address=192.168.20.0/24 routing-table=main gateway=VLAN20 immediate-gw=VLAN20 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.20.1%VLAN20

   DAc   dst-address=192.168.30.0/24 routing-table=main gateway=VLAN30 immediate-gw=VLAN30 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.30.1%VLAN30

   DIcH  dst-address=192.168.40.0/24 routing-table=main gateway=VLAN40 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.40.1%VLAN40

   DAc   dst-address=192.168.50.0/24 routing-table=main gateway=VLAN50 immediate-gw=VLAN50 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.50.1%VLAN50

   DAc   dst-address=192.168.110.0/24 routing-table=main gateway=ether1 immediate-gw=ether1 distance=0 scope=10 suppress-hw-offload=no local-address=192.168.110.254%ether1
[admin@MikroTik] > /ip firewall filter print detail
Flags: X - disabled, I - invalid; D - dynamic
 0    chain=forward action=drop src-address=192.168.60.0/24 dst-address=192.168.20.0/24

 1    chain=forward action=drop src-address=192.168.60.0/24 dst-address=192.168.30.0/24

 2    chain=forward action=drop src-address=192.168.60.0/24 dst-address=192.168.40.0/24

 3    chain=forward action=drop src-address=192.168.60.0/24 dst-address=192.168.50.0/24

 4    chain=forward action=drop src-address=192.168.60.0/24 dst-address=192.168.10.0/24
[admin@MikroTik] > /ip firewall nat print stats
Columns: CHAIN, ACTION, BYTES, PACKETS
# CHAIN   ACTION       BYTES  PACKETS
;;; Internet
0 srcnat  masquerade  254 854    1 326
[admin@MikroTik] >
```

Activate Windows
Go to Settings to activate Windows

```
[admin@MikroTik] > /tool traceroute 8.8.8.8
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
 #  ADDRESS          LOSS   SENT  LAST    AVG   BEST  WORST  STD-DEV
 1  192.168.110.1    0%        3  0.3ms   0.4   0.3   0.5    0.1
 2  192.168.0.1      0%        3  2.2ms   7.7   2.2   14.7   5.2
 3  192.168.10.1     0%        3  2.8ms   3.4   2.8   4.5    0.8
 4  192.168.100.1    0%        3  11ms    5.7   3     11     3.8
 5  10.56.0.1        0%        3  14.4ms  10.1  7.7   14.4   3
 6  161.49.2.133     66.7%     3  9.7ms   9.7   9.7   9.7    0
 7  161.49.4.244     0%        3  5.5ms   10.3  5.5   18.6   5.9
 8  161.49.1.193     0%        3  6.7ms   10.6  6.6   18.5   5.6
 9  161.49.7.196     0%        3  9.7ms   9.2   8.7   9.7    0.5
10  161.49.6.144     0%        2  10.5ms  12.5  10.5  14.4   2
11  142.250.172.244  0%        2  10.2ms  18.7  10.2  27.1   8.5
12  142.251.251.45   0%        2  9.5ms   10.2  9.5   10.8   0.7
13  142.251.244.149  0%        2  7.8ms   9.8   7.8   11.8   2
14  8.8.8.8          0%        2  12.5ms  12.4  12.3  12.5   0.1
```

```
[admin@MikroTik] > /tool sniffer quick interface=ether5
Columns: INTERFACE, TIME, NUM, DIR, SRC-MAC, DST-MAC, SRC-ADDRESS, DST-ADDRESS, PROTOCOL, SIZE, CPU
INTERFACE  TIME   NUM  DIR  SRC-MAC            DST-MAC            SRC-ADDRESS           DST-ADDRESS           PROTOCOL  SIZE  CPU
ether5     2.286  973  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    292   24
ether5     2.297  974  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    292   24
ether5     2.317  975  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    292   24
ether5     2.335  976  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.336  977  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.338  978  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.339  979  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.34   980  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.341  981  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.341  982  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.343  983  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.343  984  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    652   24
ether5     2.343  985  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.343  986  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.343  987  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.344  988  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    64    3
ether5     2.352  989  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    164   3
ether5     2.352  990  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    64    24
ether5     2.353  991  ←    14:DA:E9:66:CC:A2  FF:FF:FF:FF:FF:FF  192.168.50.200:51237  192.168.50.255:20561  ip:udp    164   3
ether5     2.354  992  →    CC:2D:E0:BB:AB:79  14:DA:E9:66:CC:A2  0.0.0.0:20561         255.255.255.255:51237 ip:udp    64    24
```

- Used ping, ARP/neighbor information, interface state, IP addressing, and route inspection to narrow down where connectivity failed.

7. Final Result / Current Status

Verified outcomes include correcting a NAT-rule issue and restoring Internet connectivity in that session. VLANs and the complete multi-device design remain configuration-dependent.

8. What I Learned

- How RouterOS interfaces, addresses, routes, NAT, and bridge VLAN tables relate.
- Why tagged/untagged port assignments and VLAN filtering must match the physical topology.
- Why changes should be made in small steps with a recovery path and before/after verification.

9. Real-World Application

These skills apply to small-office routing, departmental network segmentation, switch-port troubleshooting, and documenting a maintainable network.

10. Evidence Checklist

☑ Topology diagram with labels and IP/VLAN plan

☑ Configuration screenshot or sanitized command output

- ☑ Verification output showing successful tests
- ☑ Troubleshooting evidence: symptom → investigation → fix → retest
- ☑ Date, device/software version, and brief caption

For more information about my troubleshooting experience , you may see it to my files thank you!